



Cambridge IGCSE™

PHYSICS

0625/61

Paper 6 Alternative to Practical

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **9** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Acronyms and shorthand in the mark scheme

Acronym / shorthand	Explanation
Brackets ()	Words not explicitly needed in an answer, however if a contradictory word / phrase / unit to that in the brackets is seen the mark is not awarded.
<u>Underlining</u>	The underlined word (or a synonym) must be present for the mark to be scored. If the word is a technical scientific term, the word must be there.
/ or OR	Alternative answers any one of which gains the credit for that mark.
owtte	Or words to that effect.
ignore	Indicates either an incorrect or irrelevant point which may be disregarded, i.e., <u>not</u> treated as contradictory.
insufficient	An answer not worthy of credit <u>on its own</u> .
CON	An incorrect point which contradicts any correct point and means the mark cannot be scored.
ecf [question part]	Indicates that a candidate using an erroneous value from the stated question part must be given credit here if the erroneous value is used correctly here.
cao	Correct answer only.
ORA	Or reverse argument.

Question	Answer	Marks
1(a)(i)	10.0	1
1(b)	obtaining exact balance OR judging position of (centre of) load OR masses sliding on ruler	1
1(c)	graph: • axes right way round and correctly labelled with quantity and unit	1
	• suitable scales	1
	• all five plots correct to $\frac{1}{2}$ small square	1
	• good line judgement, thin, continuous line	1
1(d)	method shown correctly on graph	1
	candidate's value read correctly to $\frac{1}{2}$ small square	1
	value in range 25.0–29.0 cm	1
1(e)	balance ruler (on pivot with no loads)	1
	pivot point is read / balance point shows centre of mass	1

Question	Answer	Marks
2(a)	21 (°C)	1
2(b)(i)	s, °C	1
2(b)(ii)	0, 30, 60, 90, 120, 150, 180	1
2(c)(i)	$\Delta\theta = 25$	1
2(c)(ii)	$R = 0.14$ or 0.139 to 2 or 3 significant figures ONLY	1
	unit °C / s	1
2(d)(i)	continue taking temperatures / continue for a longer time (until close to room temperature) OR use lower / higher starting temperature (of water)	1
2(d)(ii)	plot a graph	1
	of temperature against time OR cooling rate against time / temperature	1
2(e)	to avoid parallax error owtte	1
2(f)	read at bottom of meniscus OR ensure measuring cylinder is on horizontal surface	1

Question	Answer	Marks
3(a)	normal at 90° and in centre of AB	1
3(b)(i)	CD is in lower left quadrant and $i = 30^\circ \pm 2^\circ$	1
3(b)(ii)	P_1 and P_2 at least 5.0 cm apart	1
3(c)(i)	straight line through pins P_3 and P_4 to AB	1
3(c)(ii)	all three lines LN , CD and CE drawn and meet or would meet at C within approx. 2 mm (by eye), drawn neatly and not too thick	1
	α in range $47^\circ - 51^\circ$	1
3(d)	any one from: - view bases of pins - place pins (as) far apart (as possible) - ensure pins are vertical - sharp pencil / thin lines / thin pins	1
3(e)(i)	β in range $70^\circ - 73^\circ$	1
3(e)(ii)	correct answer for θ (which is $90 - \beta$)	1
	correct working shown ($90 - \beta$)	1
3(f)	statement to match results with justification including the concept of within or beyond the limits of experimental accuracy	1

Question	Answer	Marks
4	MP1 circuit diagram – power supply, lamp and ammeter in series, voltmeter in suitable position to read V across lamp	1
	MP2 measure the current and potential difference	1
	MP3 calculate resistance	1
	MP4 repeat with at least two other currents / voltages / settings of variable resistor / power supply	1
	MP5 a way to change the current/voltage (e.g. adjusting / using variable resistor OR changing voltage of power supply OR adding batteries)	1
	MP6 table with columns for potential difference, current and resistance, headed with quantities and appropriate units	1
	MP7 plot graph of resistance against current OR compare values of <u>resistance</u> and <u>current</u> in table	1